

Codes	Verbs	Book Definition	Book Example (figures on subsequent slides for reference)	Scientific Team Meetings (STMs) Context
Unidirectional	builds on, draws from, responds to, influences	"Moves that link only backward [backlinks] or only forward [forelinks]" (pg 57)	(Refer Figure 4.3): Backlink: moves 4, 7, 8, 9, 12, 13, 26, 27 Forelinks: 1, 2, 5	Occurs when a participant contributes an idea that clearly builds on or responds to a previous idea, but the earlier speaker does not engage back. For example, someone proposes a biomarker, and another person adds feasibility constraints without creating a reciprocal exchange.
Bidirectional	interacts with, revisits, refines, negotiates	"Links both backward and forward." (pg 57)	(Refer Figure 4.3): 3, 6, 10, 14-25	Occurs when two participants reference each other's ideas repeatedly - such as a back-and-forth clarification or debate about whether a method works for humans versus mice. These exchanges lighten conceptual relationships and co-develop a shared understanding
Critical Moves	synthesizes, anchors, reframes, converges	"those [moves] forming a particularly large number of links." (pg 58)	(Refer Figure 4.3): Critical moves (CMs) are identified at thresholds 5, 6, and 7. At the highest level (CM 7), three moves are critical: 24, 26 (<) and 14 (>). As the threshold lowers, more moves qualify on the ">" side—moves 14, 17, and 20 at CM ⁶ , and 3, 14, 16, 17, and 20 at CM ⁵ . No "<" CMs appear at any level. Overall, 11% of moves are CM ⁷ , 19% are CM ⁶ , and 26% are CM ⁵ .	A contribution that shifts the direction of the conversation - such as a clarification that reveals a major feasibility issue, a reframing of the research question, or a synthesis that unifies multiple strands. These moments often pivot the entire group's reasoning
Chunk	interweaves, cross-references, co-develops	"A chunk is a block of links among successive moves that forms links almost exclusively among themselves and are loosely or not at all interconnected with other moves." (pg 63)	(Refer Figure 4.3): first chunk: Moves 1-14 second chunk: Moves 14-26 Note: Move 14 is a pivot move that connects the two halves ... graphically distinct as a discernible triangle called a chunk.	A chunk is a cluster of discussion where multiple participants concentrate on the same problem (e.g. biomarkers, disease mechanisms, or imaging limitations), adding refinements, challenges, and extensions. Chunks show where shared attention and co-development occur.
Web	interweaves, cross-references, co-develops	"A web is formed when a large number of links are generated among a relatively small number of moves." (pg 64)	(Refer Figure 4.3): moves 14-19 moves 20-26	This forms when many connections appear across a small set of ideas. Example: when several participants tie their expertise to the same central hypothesis, offering evidence, constraints, alternative models, and methodological angles that all interrelate.
Sawtooth Track	returns to, responds, revives, recontextualizes	"In some instances a sequence of moves link each to the preceding move. The link lines in such a case describe a zigzag pattern reminiscent of a sawtooth." (pg 65)	(Refer Figure 4.3): Between moves 22-26	Appear when participants revisit earlier ideas after exploring other topics - such as returning to the original research question or earlier mechanistic explanations after 15-20 minutes of unrelated discussion. STM conversations often loop back like this.
Link Span (L.S.)	extends across, connects over time, sustains	"The distance between two linked moves." (pg 65)	(Refer Figure 4.3): Move 6 in the ashray linkograph has backlinks to moves 3 and 5 and a forelink to move 7. For the links of move 6 to moves 5 and 7, L.S. = 1. For the link to move 3, however, L.S. = 3.	Reflects on how far apart in the meeting two related ideas appear. A short-span link might be someone immediately refining a previous idea, a long span link might be someone referencing an early-meeting assumption much later during synthesis.
Link Index (L.I.)	measures connectivity, quantifies influence	"A link index is the ratio between the number of links and the number of moves that generate them in a linkograph, or part thereof, expressed as a proportion." (pg 66)	(Refer Figure 4.3): Number of moves: 27 Number of Links: 60 $L.I._{1-27} = 2.2$	Captures how "influential" a contribution is - for example, when a participant's early framing question or conceptual hypothesis becomes referenced repeatedly throughout the meeting, generating many downstream contributions.

this is what everything feels like watching these meetings at the moment

Table 1

Verb	Meaning (plain language)	Scientific Team Meeting Example	Possible Mappings	Relevance for our Project
Builds on	Adds to a previous idea that continues or expands it	One participant describes a potential imaging biomarker, and adds supporting literature and explains how it could integrate with their lab's current work.	A short open forklift (Participant immediately adds onto someone's ideas) Part of a bigger chunk (Several build on from a cluster around the same central idea) Long open backshift (Someone revises a concept discussed 20-30 minutes ago) → wouldn't be surprised if this is relevant, unable to tell for now → though	Helps us detect how ideas accumulate toward shared hypotheses; shows where early contributive stages later scientific reasoning
Responds to	Directly replies to what someone just said - answers a question, acknowledges a point, or reacts to a suggestion	After someone raises concerns about feasibility, another member responds directly by describing how their lab overcame similar constraints	Short open backbite (Direct reply to the previous statement) Reformulated idea (Response both references and affirms the interpretation) Part of a bigger chunk (Feel seen and both becomes a cluster)	Allows us to track immediate conversational dependencies - essential for modeling how participants coordinate meaning in real time.
Clarifies	Makes a previous idea more precise, corrects a misunderstanding, or fills in missing data	Someone explains that the proposed method works in genetics but another clarifies why it wouldn't generalize to neuroscience data.	Backbite (Clarification ties back to a prior unclear move) Short open forklift (Clarifies, then addresses the idea with more detail) Revised - C&A - (establishes heavy context/move) (Many later statements dependent on this clarification)	Reveals how teams repair misunderstandings or tighten definitions, which is crucial for identifying convergence vs. confusion.
Refines	Improves or tightens a previous idea - same core idea, but made sharper, more accurate, or more workable	After a limited idea is presented, another participant refines it by specifying which biomarkers or pathways would actually be measurable in humans.	Short open forklift (Refinement improves the just prior idea) Bidirectional link (Reference both corrects backward and adds forward) Chunk evolution (Many refinements create a tight cluster of ideas/knows)	Shows how raw ideas evolve into more precise, workable scientific proposals; helps identify incremental conceptual progress.
Challenges	Pushes back on a previous idea by pointing out a limitation, flaw, or gap	A participant says representing existing drugs in a viable path, and someone else challenges the idea, arguing it produces suboptimal solutions and citing counterexamples.	Backbite with negative polarity (Pushes back on the earlier move) Long open question (Challenges something said a time ago) Critical move point (Challenge shifts the direction of the entire discussion)	Helps surface moments of productive disagreement, which our project treats as diagnostic signals for reasoning divergence.
Extends	Adds a new dimension, perspective, or possibility that goes beyond the original idea but is still connected to it.	A participant mentions studying mechanisms in mice, and another extends the idea by adding evidence from human studies and proposing a combined framework.	Short open forklift (Adds new dimensions/possibilities) Chunk membership (Someone acknowledges around a shared direction) Long open backshift (Extends something said from early on in the meeting)	Captures how teams broaden the problem space by adding new dimensions (data, methods, mechanisms), showing idea expansion over time.
Redirects	Shifts the conversation to a different topic, question, or angle - often to keep the group on track.	When discussion drifts, the facilitator redirects the group to the main question, pitching a hypothesis to move forward more quickly	Critical move (C&A) (Redirection sets a new agenda that later talk depends on) Chunk boundary marker (Shifts one conceptual region, begins another) Pivot (Entire structure reorganizes around this shift)	Helps us detect shifts in discussion trajectory and identify pivot points where the group moves toward new hypothesis or agenda items.
Grounds / Justifies	Supports an idea with evidence, data, literature, or examples from experience.	A participant grounds an abstract claim by referencing a published study, illustrating where similar dynamics occurred or failed.	Backbite (Grounding connects current claim to earlier evidence) Bidirectional link (Evidence reshapes interpretation of earlier moves) Chunk addition (Evidence anchors a cluster of ideas into a coherent structure)	Lets us measure how evidence stabilizes or legitimizes emerging ideas - a key part of understanding scientific reasoning structures.
Synthesizes	Combines multiple earlier ideas into one coherent explanation, direction, or conclusion.	Toward the end, a member synthesizes multiple viewpoints - imaging, biomarkers, causal inference - and outlines a single integrated hypothesis.	High-degree critical move (C&A - C&A - C&A) (Synthesis integrates many lines and backlinks) Chunk evolution (Central node of the whole conceptual region) Chunk-to-chunk bridge (Connects two previously separate strands)	Points to convergence moments where multiple idea threads unify; helps validate whether and how teams reach shared understanding.

Table 2

Verb	Relevant STM Codes	How to Detect it in Data	Linkographic Meaning	Relevance to Project
Builds on	Idea Management; Knowledge Sharing; Integration Practices	Occurs after another idea (Idea Management = +1 or 0); Speaker references previous content ("adding on...", "in addition...", "building from what X said..."); Temporal adjacency: same idea cluster within ~1-2 turns	A forelink from on move to the next where the second move depends on or extends the content of the first. "B" → "A" where B incorporates part of A's idea - Classic short-span forelink	Shows how ideas accumulate. Tracking "builds on" helps identify idea growth chains, where multiple participants progressively elaborate the same scientific direction. Useful for reconstructing the trajectory of candidate proposals.
Responds to	Information Seeking; Knowledge Sharing; Participation Dynamics	Follows a question (Information Seeking = +1); Direct lexical reference to prior turn ("yes", "to answer...", "regarding that..."); Short temporal gap → immediate reply	A backlink, where a later move addresses, answers, or reacts to a previous move. "B" → "A" but the purpose is acknowledgement / direct response.	Captures conversational dependency. Helps reveal how question, gaps, or uncertainties trigger specific clarifications or shifts, allowing us to map how the team maintains coherence and repairs misunderstandings.
Clarifies	Knowledge Sharing; Evaluation Practices; Integration Practices	Knowledge Sharing code = 1 or 2 following confusion/doubt; Often triggered by someone expressing uncertainty ("I'm not sure...", "Do you mean...?"); Provides missing detail or correction that resolves ambiguity	A backlink-heavy move that resolves confusion, refines definition, or anchors an earlier ambiguous move Often forms potential <CM> (critical move due to many backlinks)	Clarification stabilizes shared understanding. These moments show where ambiguity is reduced, enabling later ideas to rest on firmer ground. Important for identifying foundational moves the model must recognize to predict convergence.
Refines	Idea Management; Knowledge Sharing; Evaluation Practices	Evaluation Practices = 1 or 2 (but constructive, not negative); New move comes after a board or vague idea and adds specificity; Language like "more precisely...", "a better framing might be..."	A forelink that narrows, improves, sharpens, or constrains an earlier concept - Usually part of a cluster of related forelinks - Contributes to shaping the design constraint	A refinement narrows or sharpens a concept. These moments help us trace how broad hypotheses evolve into more precise, testable versions, showing mid-stage conceptual evolution in scientific ideation.
Challenges	Evaluation Practices; Relational Climate; Participation Dynamics	Evaluation Practices = -1 or 0 (critical tone); Relational Climate may dip (sarcasm, sharp tone) Keywords: "I disagree", "this won't work", "but the issue is..."	A long-span backlink that contradicts or destabilizes a prior move - often triggering a redirection of the design path. - Can create branching in linkograph structure - Seed of a pivot in the design trajectory	Challenge moves reveal scientific tension points - where assumptions break, methods are questioned, or the group changes direction. Critical for detecting pivots and understanding how disagreement reshapes the scientific trajectory.
Extends	Idea Management; Knowledge Sharing; Integration Practices	Idea Management = +1 for elaboration; Knowledge Sharing = 1 or 2 with new domain evidence; Pivots outward: "Another angle...", "Additionally...", "We could also consider..."	A forelink that adds new dimensions or information to an earlier move - Grows the design graph outward - Often relevant to divergent expansion	Extension pushes an idea outward into new territory. These moves identify explanatory reasoning, showing where the group begins generating alternative directions or adjacent research paths.
Redirects	Coordination & Decision Practices; Participation Dynamics; Idea Management	Coordination & Decision Practices = 1 or 2; Speaker shifts scope ("Let's get back to...", "Maybe we should move to..."); Participation Dynamics change (facilitator stepping in)	A move that alters the direction of the design trajectory, often through multiple backlinks + an initiating forelink. - Can form a critical move if many subsequent ideas depend on it. - Linkography sees this as a pivot node	A redirect captures intentional course correction - either back to the main research question or toward a more promising scientific angle. These moves reveal facilitator and leadership influence on idea flow and meeting structure.
Grounds/Justifies	Knowledge Sharing; Evaluation Practices; Integration Practices	Knowledge Sharing = 2 (accurate evidence); Citations of studies, lab results, or methods; Language: "In our data...", "According to X study...", "Evidence shows..."	A backlink that introduces evidence, constraints, or reasoning to support an earlier move - Stabilizes an idea in the linkograph - Often forms the root of a stable cluster	Grounding connects a claim to empirical evidence, prior work, or methodological constraints. Shows how the team anchors abstract ideas into feasibility, which is essential for distinguishing speculation from actionable proposal seeds.
Synthesizes	Integration Practices; Idea Management; Coordination & Decision Practices	Integration Practices = 2 (balanced, coherent summary) Speaker references multiple prior moves ("So far we've...", "Putting these together..."); Often appears near decision points or group alignment	A multi-directional move that pulls together several earlier moves (multiple backlinks) and creates a unifying idea (forelink) - Often marked as <CM> (critical in BOTH directions) - Connects clusters and indicates moment of creative integration	Synthesis integrates multiple partial ideas into a unified direction. These are high-value conceptual consolidation moments - often the closest thing to "proposal crystallization" visible in early-stage ideation.

Table 3